

# ANNISA RACHMAN

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## RESEARCH INTERESTS

Personalised Medicine, Dosimetry, Nuclear Medicine Alpha and Beta Therapy, PBPK, Bioengineering, Cancer Therapeutics and Diagnostics, Drug Design and Development, Medical Physics, Translational Science, Pharmacokinetics (PK) and Pharmacodynamics (PD), Biokinetics Imaging Studies, AI for Nuclear Medicine, Biomarkers.

## EDUCATION

(Incoming Student) **Johns Hopkins University, School of Medicine: MSc in Medical Physics** **Intake Aug 2026**

University ID: C05DB5

- Merit Scholarship Awardee from Johns Hopkins University School of Medicine
- LPDP World Leading Universities (PTUD) Fully Funded Scholarship Awardee from Indonesian Government

**University of Tsukuba: Summer Research Program (Medicine)**

**Sep 2025**

Grade: 100/100

**Universitas Indonesia: BSc in Physics (Medical Physics and Biophysics Concentration)**

**Sep 2020 – Jan 2024**

Grade: GPA 3.62/4.00 with *Cum Laude* honors

- First Class (1<sup>st</sup>) in Year 4 with an average GPA of 3.95/4.00
- First Class (1<sup>st</sup>) in Year 3 with an average GPA of 3.75/4.00

Academic Achievements:

- The "Best Graduate" of the Physics Department, Universitas Indonesia, Graduation Ceremony in March 2024
- The "Fastest Graduate" and the "First Student" in the History of the Nuclear Medicine Research Group at the Universitas Indonesia to Complete a 4-year Bachelor Program in just 3 years and 4 months, an Achievement Reached by only 3 out of 120 Students in the Physics Department
- Finalist (Top 6) for The Most Outstanding Student of the Faculty of Mathematics and Natural Sciences Universitas Indonesia 2023, Representative Among 5,000+ Students
- The Most Outstanding Student of Physics Universitas Indonesia 2023, Representative Among 480+ Students
- Obtained 3 Intellectual Property Rights (IPRs) in Indonesia
- Secured Undergraduate Scholarship (~\$3050), Fully Funded by the Ministry of Education, Culture, Research and Technology of the Republic of Indonesia

Relevant Courses:

*Medical Imaging & Nuclear Medicine, Radiological Physics and Dosimetry, Biophysics, Biomaterials, Radiotherapy Physics, Materials Characterization Method, Anatomy & Physiology, Statistical Physics, Physical Computing, Nuclear Physics, Health Physics & Radiation Protection, AI, (Capita Selecta 1) Machine Learning, Electronics.*

**Bandung Institute of Technology: International Modern Optics and Applications (SCoMOA)**

**Jul – Aug 2023**

Grade: First Class (1<sup>st</sup>) with 79.69% equivalent to GPA: 4.00/4.00

**Universitas Indonesia: Exchange Program on Computer Science: Parallel Programming**

**Sep – Dec 2021**

Grade: First Class (1<sup>st</sup>) with 95.62% equivalent to GPA: 4.00/4.00

**Gadjah Mada University: Exchange Program on Medical Physics**

**Aug – Dec 2021**

Grade: First Class (1<sup>st</sup>) with GPA: 4.00/4.00

## UNDERGRADUATE THESIS

*"Study on the Effect of Model Selection on Absorbed Dose Calculation Radiation in the Case of Animal to Human Biokinetic Translation"*

This study focused on Nuclear Medicine Dosimetry, PK/PD modeling on clinical translational research with biokinetics data <sup>177</sup>Lu-OPS201 radiopharmaceutical from animals (pigs and mice) to humans (patients with Neuroendocrine Tumours (NETs)). MIRD concepts, the SoE model, GoF analysis, and AICc were used for model selection methods to evaluate the efficacy and safety of clinical translational research with allometric scales, time scales, and mass scales affect the accuracy and precision of the *Time-Integrated Activity Coefficient* (TIAC), which directs to absorbed doses.

Advisors : Dr. Deni Hardiansyah (Universitas Indonesia)  
Nur Rahmah Hidayati, MSc (Indonesia's Nuclear Research Agency)

Examiners : Prof. Supriyanto Ardjo Pawiro (Universitas Indonesia)  
Dr. Syahril Siregar (Universitas Indonesia)

*\*This work has been presented to 50+ students and archived at the Universitas Indonesia Library.*

## SKILLS

**Programming and Software:** SAAM II, OLINDA/EXM, NONMEM, Python, MATLAB, R, LaTeX, SQL, MySQL, VHDL, Oracle VM, Vivado, HTML, EasyEDA, Multisim, Proteus, GNU Plot, GraphPad Prism, Google Collab, VS Code.

**Physics Lab:** Nuclear Magnetic Resonance (NMR), Cs/Ba-137m Isotope Generator, Geiger-Muller counter, Gamma Spectroscopy, Thermal Radiation, Digital ICs, Arduino, FPGA, etc.

**Languages:** C1 English, A1 German, A1 Arabic, Native Indonesian, Native Sundanese.

## RESEARCH EXPERIENCES

**Research Assistant**, lab of Dr. Deni Hardiansyah (Nuclear Medicine Lab) **Sep 2024 – Present**  
Medical Physics and Biophysics, Universitas Indonesia (Depok, Indonesia)

- Drafting research proposals and preparing funding applications to support ongoing and future research in the lab.
- Collaborating with PhD students, 3 academic faculties, and 7 Indonesian researchers on Nuclear Medicine projects.
- Guiding and mentoring 2 undergraduate students in their research thesis projects.
- Creating virtual data patients and evaluating Machine Learning models for population-based model selection dosimetry method using MATLAB.

**Research Assistant**, Nuclear Medicine Group, Indonesia's Nuclear Research Agency (BRIN) **Jan 2025 – Present**  
Research Centre for Nuclear Safety, Metrology and Quality Technology.

- Finding the best model using model selection methods for data biokinetics  $^{177}\text{Lu}$ -EDTMP using SAAM II.
- Conceptualizing the best Time-Points combinations for Simplified Dosimetry using Python.
- Collaborating with 2 Indonesian researchers: Nur Rahmah Hidayati and Bisma Barron Patrianesha.

**Medical Physicist/Research Assistant**, group of Dr. Lukmanda Evan Lubis **Sep 2024 – Mar 2025**  
Department of Diagnostic and Interventional Radiology, Universitas Indonesia Hospital (RSUI) (Depok, Indonesia)

- Assisting in the operation and maintenance of diagnostic imaging equipment, conducting QA tests and calibration.
- Contributing to industrial research and clinical projects on imaging techniques and safety standards.
- Calculating SDNR for phantom analysis and comparing with ICRP and national guidelines.

**Writer and Illustrator for Nuclear Medicine Book**, Nuclear Medicine Lab **Jan 2024 – Present**  
Medical Physics and Biophysics, Universitas Indonesia (Depok, Indonesia)

- Contributing to the First Indonesian Nuclear Medicine Physicist Project.
- Assisting as the 3<sup>rd</sup> Author of the book project with Dr. Deni Hardiansyah and Intan Apriliani S.M., MSc.
- Creating 15+ illustrations using Inskap and Adobe to enhance the visual and educational value of the book.

**Undergraduate Student Researcher**, lab of Nur Rahmah Hidayati, MSc (Nuclear Medicine Lab) **Jul – Dec 2023**  
Indonesia's Nuclear Research Agency (BRIN) (Jakarta, Indonesia)

- Discovered areas in radiation dose calculations (dosimetry) using translational (mass, time, and allometric scaling) and model selection methods (GoF and AICc).
- Conducted research on SAAM II, MATLAB, and Python for Nuclear Medicine dosimetry applications and analysed data from  $^{177}\text{Lu}$ -OPS201 radiopharmaceutical and Neuroendocrine Tumours (NETs) patients.
- Visualized Time-Activity Curves (TACs) in SAAM II and GraphPrism.

**Undergraduate Student Researcher**, lab of Dr. Deni Hardiansyah (Nuclear Medicine Lab) **Jan – Dec 2023**  
Medical Physics and Biophysics, Universitas Indonesia (Depok, Indonesia)

- Identified accuracy and precision for Nuclear Medicine dosimetry calculation and nuclear safety for radiopharmaceuticals  $^{177}\text{Lu}$ -OPS201 using SAAM II, Python, and MATLAB.
- Executed PK/PD computational modeling, dosimetry parameters analysis, and radiopharmaceutical (RPT) imaging studies for animals (mice and pigs) and humans (NETs patients) using a population-based model selection method.
- Performed TIAC (Time-Integrated Activity Coefficient) calculations and developed translational models.
- Discovered that the best model with average human biokinetic data can predict TIAC of  $5.45 \pm 0.03$  hours with 99.20% accuracy close to the TIAC reference.

**Research Intern**, lab of Nur Rahmah Hidayati, MSc (Nuclear Medicine Lab) **Jan – Jun 2023**  
Indonesia's Nuclear Research Agency (BRIN) (Jakarta, Indonesia)

- Observed Internal Dosimetry Nuclear Medicine Therapy, used MIRD with time correction to estimate the total organ effective dose in Thyroid on males and females with various injections.
- Implemented research on biodistribution data using OLINDA and Python for radionuclide I-131 to obtain effective absorbed doses and compared with the ICRP 128 reference for human radiation dose safety limits in Thyroid organs.

## TEACHING EXPERIENCES

**Teaching Assistant** for 7 Undergraduate Class Courses, Universitas Indonesia **Jan 2022 – Dec 2023**

- SCPH603135: *Introduction to Nuclear Physics*, class of Prof. Anto Sulaksono (39 students), 2023
- SCPH603716: *Introduction to Radiological Physics and Dosimeters*, class of Dr. Akbar Azzi, Dr. Lukmanda Evan Lubis, and Siti Julia, MSc (23 students), 2023
- SCPH602223: *Classical Mechanics*, class of Prof. Anto Sulaksono (31 students), 2023
- SCPH602221: *Electromagnetic Fields I*, class of Dr. Budhy Kurniawan (30 students), 2023
- SCPH602133: *Modern Physics*, class of Prof. Supriyanto Ardjo Pawiro (32 students), 2022
- SCPH602156: *Electronics II*, class of Dr. Djati Handoko (25 students), 2022
- SCPH601110: *Basic Physics for Mathematics Students*, class of Surya Darma, MSc (60 students), 2022

**Laboratory Assistant** for 4 Undergraduate Laboratory Courses, Universitas Indonesia **Aug 2022 – Dec 2023**

- SCPH601142: *Basic Physics for Engineering Students*, assisted 152 students, 2023
- SCPH602246: *Advanced Physics for Physics Students* (NMR module), assisted 64 students, 2023
- SCPH602147: *Electronics II for Physics Students*, assisted 34 students, 2023
- SCPH601142: *Basic Physics for Physics Students*, assisted 96 students, 2022

## AWARDS AND HONOURS

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- 2025 **LPDP Scholarship (Fully Funded)**, Indonesian Government for World Leading Universities (PTUD).
- 2025 **Merit Scholarship**, Johns Hopkins University School of Medicine.
- 2025 **2nd Best AI Innovators Indonesia**, AI Talent Hub Indonesia.
- 2025 **Selected Mentee for Indonesia Mengglobal**, Indonesia Mengglobal Community.
- 2025 **Selected Poster Presenter**, The 27th Annual Scientific Meeting, Indonesian Society of Nuclear Medicine.
- 2025 **Summer Research Program Institute of Medicine**, University of Tsukuba, Japan.
- 2025 **Top 100 Best Innovators AI Talent Hub Indonesia**, AI Talent Hub Indonesia X UI Innovation.
- 2025 **Selected Start-up for Problem Solution Fit Funding (SOFIAA by NuMTech ID)**, Hackathon UI Incubate.
- 2025 **Finalist Indonesia's delegate (Special Funded)** for the Global Education Türkiye 2026 Chapter 3.
- 2024 **Indonesia's delegate** for the Bali International Student Festival (BISF) in the Education Panel.
- 2024 **Invited speaker**, "Writing Class with Natural Sciences Strategic Association Universitas Indonesia".
- 2024 **Invited speaker**, "Time Management for Studying and Organisation", Impact Class BMKA Salman ITB.
- 2024 **Honored** with the 2<sup>nd</sup> Most Outstanding Salman Activist Award 2024 in the Academic Category, ranked 2<sup>nd</sup> among 39,459 student activists from 743 universities for academic excellence and leadership.
- 2024 **Most Outstanding Graduate** at the Salman ITB April Graduation Celebration Event.
- 2024 **Delivered** a commencement speech at the graduation ceremony of the Physics Universitas Indonesia 2024
- 2023 **Most Ambitious Student** in the Salman Activist Scholarship at Leading Figure Fest Award, recognized among 60 selected outstanding undergraduate students in Indonesia.
- 2023 **Awarded** the "Best Group Award of International Youth Leader Conference and Culture Exchange in Malaysia", surpassing 35 delegates with an impactful presentation.
- 2023 **Top 10 recipients** of the Shape Your Life Bootcamp, earning a scholarship for an intensive educational development program out of 200+ applicants.
- 2023 **Awarded** Merit Scholarship of ASEAN Global Youth Learning Exchange & Excursions, facilitating an Exchange Program across Malaysia, Thailand, and Singapore.
- 2023 **Awarded** the Aspire Leaders Program by Aspire Institute at Harvard University.
- 2023 **Indonesia's delegate** for the International Youth Leader Program Batch 49 Chapter Kuala Lumpur, Malaysia, emerged successfully from 20,000+ applicants.
- 2023 **Universitas Indonesia's student** for the International Summer Course on Modern Optics and Its Application (SCoMOA) held by the Indonesian Optical Society (InOS), in collaboration with Institut Teknologi Bandung, Universitas Pelita Harapan, Universitas Padjadjaran, and the University of Malaya.
- 2022 **Awarded** the Activist Salman Scholarship, selected from 1,082 applicants.
- 2022 **Observer participant** at International Summer Course on Medical Physics, Universitas Diponegoro.
- 2022 **Honorable delegate** at the Asia Global Students Summit Conference & Exchange Program, earning recognition among 60+ international delegates.
- 2022 **Best team** at the Asia Global Students Summit Conference & Exchange Program.
- 2022 **Indonesia's delegate** at the Asia Global Students Summit Conference & Exchange with IIUM (Malaysia), Tokyo International University, ERC Institute Singapore, and Konkuk University South Korea.
- 2021 **Ambassador** Asia Youth Summit, Youth Break the Boundaries
- 2020 **Indonesia's delegate** for Sustainable Development Goals No.4 (Quality Education) at the Asia Youth Summit.
- 2018 **Finalist** of the National Science Writing Competition "AEROCREATION" at the Faculty of Mechanical and Aerospace Engineering of Institut Teknologi Bandung (ITB), ROHEA: pioneering innovative flight mechanisms.

## PUBLICATIONS

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**Rachman A.**, Ardjo Pawiro S., Hardiansyah D., Hidayati N.R., Nuraeni N., Patrianesha B.B. (2025). *Improving the Radiation Protection of Patients in Biomedical Research through Better TIAC Prediction for New Radiopharmaceuticals*. International Atomic Energy Agency (IAEA). International Conference on Radiation Protection in Medicine: X-Ray Vision. Abstract ID: 268.

[In process] Hardiansyah D., Mu'minah I.A.S., and **Rachman A.** (2024). *Nuclear Medicine Physics Book*, Indonesia.

**Rachman A.**, Hardiansyah D., Hidayati N.R. (2024). *Study on the Effect of Model Selection on Absorbed Dose Calculation Radiation in the Case of Animal to Human Biokinetic Translation*. Universitas Indonesia. No. Barkod 14-24-39060873.

**Rachman, A.** (2021). "Sowing Seeds of Kindness". *Fight or Flight*, edited by Andi [et al.], KMO Indonesia, pp. 59. ISBN: 978-623-332-143-3.

## SEMINARS AND CONFERENCES

- International Conference on Radiation Protection in Medicine: X-Ray Vision** 8 – 12 December 2025  
“Improving the Radiation Protection of Patients in Biomedical Research through Better TIAC Prediction for New Radiopharmaceuticals”  
Represented Indonesia in the International Atomic Energy Agency (IAEA), Austria.
- Bali International Student Festival (BISF) by Indonesia Youth Foundation** 26 Sep – 29 Sep 2024  
“Empowering Artificial Intelligence (AI) in Education”  
Represented 30 education delegates and delivered presentations to 100+ international delegates from diverse fields.
- International Youth Leader Conference and Culture Exchange** 28 May – 01 Jun 2023  
“Leadership Styles and Their Influence on Youth Academic Performance: An Islamic Perspective in Indonesian Youth”  
Presented as one of 35+ international delegates on education topics at a conference attended by over 500 people.
- Seminar Finalist Most Outstanding Student Faculty of Mathematics and Natural Sciences** 10 Mar 2023  
“RT (Robot Translator): Improving the Quality of Education for Hearing Loss Children in Rural Areas”  
Presented the concept idea and development plan to 30+ experts and outstanding students.
- Asia Global Students Summit Conference & Exchange Program** 15 – 21 Aug 2022  
“Getting to Know the German Environment and Potential Collaborations with Asia”  
Represented the group in presenting on environmental topics, specifically climate change, to 60+ delegates.
- Asia Youth Summit Conference by Youth Break the Boundaries** 11 – 12 Dec 2020  
“Increasing and Equalizing the Quality of Collegiate Education in Indonesia”  
Indonesia’s delegate on education topics among 150 international delegates, ambassador for the next year’s program.

## ADDITIONAL EXPERIENCES

- Assistant Secretary General**, Academic Senate Universitas Indonesia (SA UI) Nov 2024 – Present  
Documenting meetings with UI’s Rectorate and executive stakeholders while supporting Academic Senate operations.
- Chief Operating Officer (COO)**, Nuclear Medicine Technology for Indonesia (NuMTech ID) Jul 2025 – Present  
Led a company for strategic planning, operational management, website developer, and partnership development to advance nuclear medicine technology in Indonesia.
- Head of Research Division**, Natural Sciences Strategic Association Universitas Indonesia Nov 2020 – Feb 2023  
Led a team of 14 members across 7 projects within the Research Division.
- Administrative Staff**, Medical Physics Summit Universitas Indonesia Jan – Mar 2023  
Managed administrative tasks, including registration and correspondence to committees, participants, and invitees.
- CTO and Science Tutor of Human Resources House (Rumah SDM)**, Indonesia Jan – Dec 2023  
• 3<sup>rd</sup> Best National Project Award and secured funding ~\$380 (Rp5,700,000).  
• Community service project with 23 volunteers to improve the quality of education of 65+ street children.
- Project Leader**, Community Service Research Project with Dr. Anne Meylani Jan 2022 – Feb 2023  
Centre Independent Learning, Universitas Indonesia  
Obtained 3 IPRs and secured funding of around ~\$660 (Rp10,000,000). Program to improve knowledge of the Indonesian National History of 60+ Residents of the Darus Shalihin Orphanage in Bogor through the Museum Visit.
- Data Scientist Intern**, Id/x Partners, Indonesia Jul 2022  
Developed data pre-processing for predictive analytics to optimize model accuracy with MySQL and Python.
- Big Data Analyst Intern**, PT. Kimia Farma, Indonesia Jun 2022  
Processed big data sets using Python and MySQL and performed visualizations using Google Data Studio.
- CEO and Founder of Physics Learning House (Rumah Belajar Fisika)**, Indonesia Oct 2020 – Feb 2021  
• Community service project to help Indonesian students improve their understanding of Physics.  
• Taught 33 Indonesian students, including 19 undergraduate and 14 senior high school students.
- Website Developer**, Physics Department Orientation 2021, Universitas Indonesia Jul – Oct 2021  
Processed big data sets using Python and MySQL and performed visualizations using Google Data Studio.
- Volunteer and Leader for 30+ events**, Indonesia 2020 – 2024

## LICENSES AND CERTIFICATIONS

- IELTS, C1, 2025
- IBM, *Code Generation and Optimization Using IBM Granite*, 2025
- FreeCodeCamp, *Data Analysis with Python*, 2024
- Great Learning, *Data Visualization With Power BI*, 2024
- Great Learning, *Data Visualization With Tableau*, 2024

**European Commission**, *Digital Skills Self-Assessment Level 6 out of 6*, 2022  
**Dicoding Indonesia – Google Developers Training Partner**, *Programming with Python*, 2022  
**International Atomic Energy Agency (IAEA)**, *Basic Training Course on IAEA Safeguards*, 2022  
**DQLab**, *Introduction to Data Science with Python*, 2022  
**DQLab**, *Introduction to Data Science with R*, 2022  
**Coursera – Yale University**, *Anatomy of the Chest, Neck, Abdomen, and Pelvis*, 2022  
**Coursera – Lund University**, *Medical Applications of Particle Accelerators*, 2022  
**Universitas Indonesia**, *Bio Risk Training in the Laboratory*, 2021  
**Universitas Indonesia**, *Chemical Safety Training in Laboratory*, 2021  
**Universitas Indonesia**, *Basic Health Training, Safety, Security, and Environment in the Laboratory*, 2021  
**Google**, *Google Analytics for Beginners*, 2020

## **REFERENCES (\*8 Available Upon Request)**

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